# VTX User Manual

Exploring and editing a procedural universe with the Virtual Terrain Explorer

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# Part I: User guide

## Introduction

VTX (the Virtual Terrain Explorer) is a program for exploring a procedurally generated universe, from whole galaxies down to the pebbles on a planet's surface. Nothing in a VTX universe is stored as a model or a map. Planets, clouds, rocks and plants are computed from short expressions while you watch, and each one gets more detailed as you get closer.

You can use VTX in three ways:

- **Explore.** Fly between stars, orbit planets and walk across their surfaces.
- **Edit.** Change any object through the Properties dialog, from the height of a mountain range to the color of a sunset, and see the result immediately.
- **Create.** Generate new stars, planets and moons with a single command, then refine them and save them as scenes or reusable objects.

This manual explains how to operate VTX. Part I is a guide to installing the program and to the tasks you will do most often. Part II is a reference to every menu, key and dialog control. Part II is generated from the program's source code, so it always matches the version of VTX it ships with.

How VTX works internally is covered in the separate VTX Theory of Operation document.

### *Conventions*

- Menu commands are written as **File > Open**.
- Keys are written as they are typed. VTX's keyboard commands are case-sensitive, so **s** and **S** are different commands.
- Values with units in the dialogs use the units shown in the control's label, for example *Day(hrs)* or *Ht(mls)* (miles).

# Installing and starting VTX

## System requirements

- Windows 10 or 11, 64-bit.
- A graphics card and driver with OpenGL support for GLSL and geometry shaders. A dedicated graphics card is strongly recommended.
- About 1 GB of free disk space, more if you generate many worlds or record movies.

## Installing

VTX is distributed in two forms:

- **Installer** (*VTX-win64-version-setup.exe*). Run it and follow the prompts. VTX is installed for the current user in *%LOCALAPPDATA%\Programs\VTX*, and no administrator rights are needed. The installer adds VTX to the Start menu, optionally to the desktop, and to Settings > Apps for uninstalling.
- **Zip archive** (*VTX-win64-version.zip*). Unpack it into any folder you can write to, for example your Documents or Downloads folder.

> **Important:** VTX writes into its own folders while it runs: generated shader files, saved scenes, preferences and generated textures. Don't install or unpack it under *C:\Program Files*, where ordinary users can't write.

## Starting VTX

Start VTX from the Start menu entry or desktop icon (installer), or by double-clicking **VTX.bat** in the VTX folder (zip). Don't start *bin\vtx.exe* directly from Explorer: VTX must start with the *bin* folder as its working directory, and the shortcut and VTX.bat take care of that.

When VTX quits it saves your session as *Saves\last.spx*. The next time you start it the same way, it reopens that session. The first time, it opens the default scene instead.

To pin VTX to the taskbar when using the zip version, create a shortcut whose target is `C:\Windows\System32\cmd.exe /c "C:\path\to\VTX\VTX.bat"`, set its **Run** option to *Minimized* and its icon to *bin\vtx.exe*, then right-click it and choose **Pin to taskbar**.

## Command-line options

VTX.bat passes its arguments to vtx.exe:

| Argument | Effect |
|---|---|
| `--s`, `--m`, `--b` | Small, medium (default) or big window |
| `--w`, `--p` | Wide or panoramic window |

| Argument | Effect |
|---|---|
| *name* | Open scene *name* from the Saves folder, for example `alpine` |
| `-`*key* | Apply a single-key command at startup, for example `-s` toggles shadows |

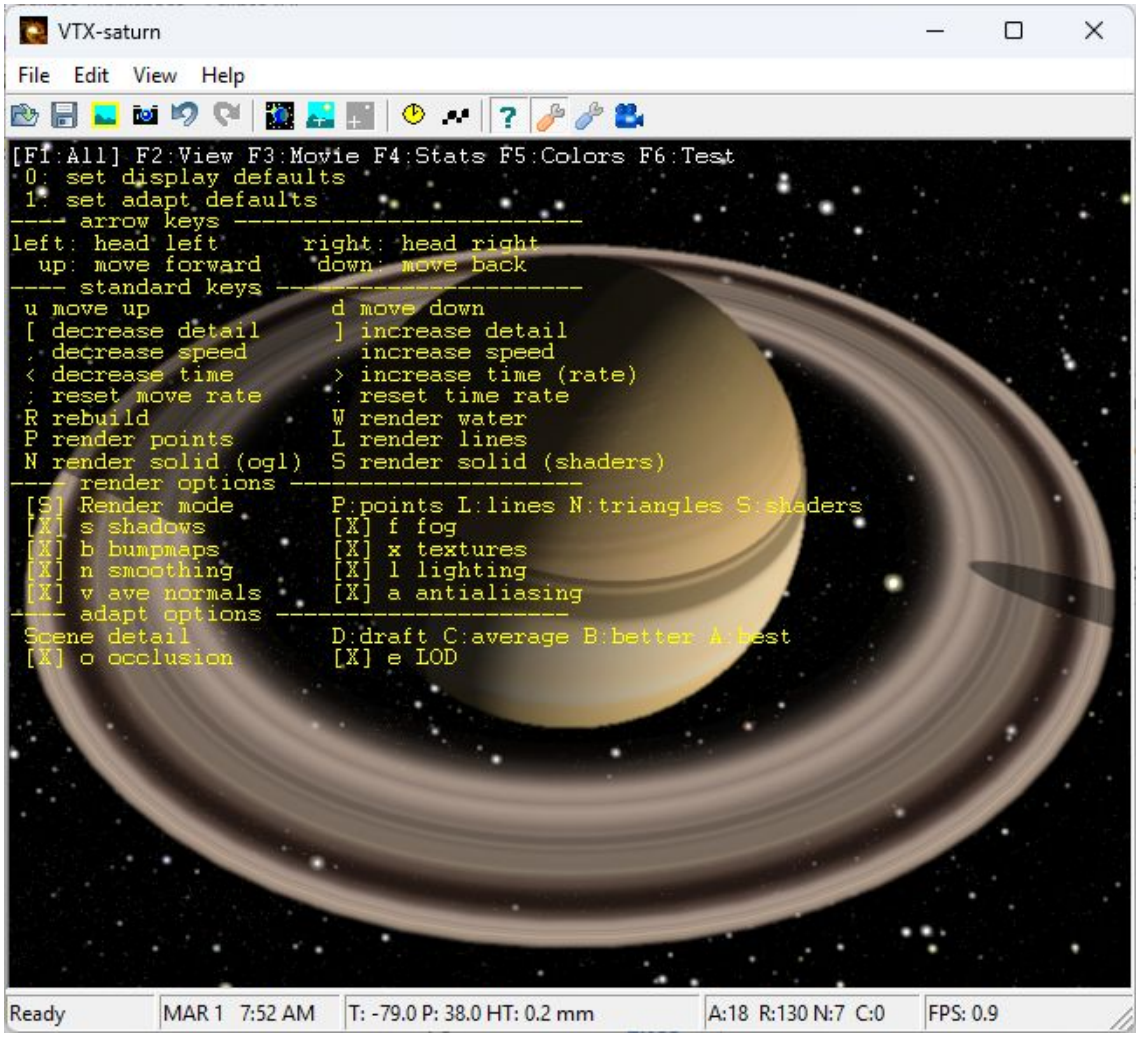
## The VTX folder

| Folder | Contents |
|---|---|
| bin | The program |
| Docs | This manual and the Theory of Operation |
| Saves | Scene files (*.spx*), including *last.spx* and *Default.spx* |
| Objects | Saved objects, by type (a new installation includes only plants) |
| Textures | Image textures, plant images and sprite images |
| Bitmaps | Textures that VTX generates for new worlds (see Generating new worlds) |
| Images | Snapshots saved with **File > Save Image** |
| Movies | Movie frames |
| Resources | Star and cloud sprites, and saved preferences |
| Shaders | The program's GPU shaders |

## Graphics card selection

Many laptops have two graphics processors: a power-saving one built into the processor and a faster dedicated one. VTX asks Windows for the dedicated graphics card. If pictures look coarse or slow, check which one is in use: VTX prints it in its console window when it starts, on a line beginning *OpenGL:*. You can also set it in **Settings > System > Display > Graphics**: add *bin\vtx.exe* and choose **High performance**.

# The main window

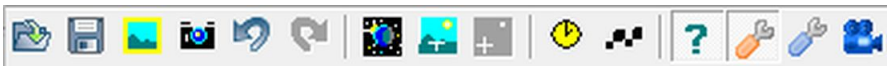



*The main window, with the on-screen key help (F1) showing*

The main window shows the scene, with a menu bar and toolbar above it and a status bar below it.

## *The toolbar*

The toolbar gives one-click access to the commands you use most. Rest the pointer on a button to see its name and what it does. This manual refers to the buttons by the names below.



*The toolbar, with Key Help on. Redo and Selection are grayed out because there is nothing to redo or select.*

From left to right, the buttons fall into four groups: files and view history, moving, animation, and help and dialogs.

| | Name | What it does | Same as |
|---|---|---|---|
|  | **Open** | Opens a scene file (.spx) and builds it. Scenes are kept in the *Saves* folder. | File > Open |
|  | **Save** | Saves the current scene, including changes made in the Properties dialog, to a scene file (.spx). | File > Save |
|  | **View Image** | Shows a saved JPEG or BMP image, such as a snapshot, in the main window. | File > View Image |
|  | **Snapshot** | Saves the current view as a JPEG or BMP image. You choose the file name; the *Images* folder is the default. | File > Save Image |
|  | **Undo** | Goes back to the previous view in the view history. Grayed out when there is no earlier view. | Edit > Undo, **-** |
|  | **Redo** | Goes forward to the next view in the history, after an Undo. Grayed out when there is no later view. | Edit > Redo, **+** |
|  | **Orbit** | Moves out to an orbital view of the current object, so you see the whole world. With a star selected, takes you to that star's system. | View > Move to > Orbit, **Shift+U** |
|  | **Surface** | Moves down to the surface of the current object. If you have marked a point with the white crosshair, takes you there. | View > Move to > Surface, **Shift+G** |
|  | **Selection** | Moves to the point marked with the white crosshair, keeping your current height. Grayed out when nothing is selected. | View > Move to > Selection, **Shift+M** |
|  | **Time** | Starts or stops time: planets rotate and orbit and clouds drift. The button stays pressed while time runs. | View > Animate > Time |
|  | **Motion** | Starts or stops motion: the view keeps moving at the current speed. The button stays pressed while it is on. | View > Animate > Motion |
|  | **Key Help** | Shows or hides the list of keyboard commands over the scene. The button is pressed while the list shows. | Help > key help, **?** |
|  | **Properties** | Opens the Properties dialog, where you inspect and edit the objects in the scene. | **Tab** |
|  | **Image Editor** | Opens the Image Editor, for creating and editing the images used as textures. | |
|  | **Movie Editor** | Opens the Movie Editor, for recording and playing paths and movies. | |

Buttons that turn something on, such as **Time**, **Key Help** and the three dialog buttons, stay pressed while that thing is on or open. Click again to turn it off or close it. **View > Show Toolbar** hides and shows the toolbar.

### *On-screen information and key help*

VTX draws status information over the scene. Press **?** (or click the **Key Help** toolbar button) to show or hide a list of the keyboard commands available in the current mode.

The function keys **F1** to **F6** switch between modes. Each mode gives the number keys **0** to **9** a different meaning:

| Key | Mode | Number keys |
|---|---|---|
| F1 | All (default) | Display and detail defaults |
| F2 | View | Turn, look and tilt the view |
| F3 | Movie | Record and play paths and movies |
| F4 | Stats | Show statistics about objects, maps and rendering |
| F5 | Colors | Color the terrain by internal properties (diagnostic) |
| F6 | Test | Alternative views for testing |

The full list for each mode is in Part II, Keyboard.

# Moving around

### *View types*

VTX has three ways of moving, which you choose with **View > Move to**, the toolbar or the keyboard:

- **Orbit** (**Shift+U**). The view circles the current object, typically a planet, at a distance. Use it to look at a whole world. If a star is selected, Orbit takes you to that star's system instead (see Traveling to the stars).
- **Surface** (**Shift+G**). The view moves down to the surface of the current object, as if you were walking or flying above the ground. If you have marked a point with the crosshair, Surface takes you there.
- **Selection** (**Shift+M**). The view moves to the point marked with the crosshair and keeps its current height.

### *Keyboard movement*

| Key | Action |
|---|---|
| Up / Down arrow | Move forward / back |
| Left / Right arrow | Turn left / right |

| Key | Action |
|---|---|
| u / d | Move up / down |
| Ctrl+u / Ctrl+d (also Ctrl+Up / Ctrl+Down) | Tilt the view up / down |
| , / . | Decrease / increase the speed of movement |
| ; | Reset the speed of movement |

In surface view, the F2 (View) mode adds keys to look and turn in fixed steps, and **9** and **0** switch between an overhead view and a normal surface view.

### Mouse movement

| Mouse | Orbit view | Surface view |
|---|---|---|
| Left-drag sideways | Circle around the object | Turn |
| Left-drag up or down | Move closer or farther | Move forward or back |
| Right-drag sideways | Circle around the object | Look to the side |
| Right-drag up or down | Tilt the view | Look up or down |
| Wheel | Zoom in or out | Lower or raise the viewpoint |
| Hold middle button | Fly forward while held | Fly forward while held |

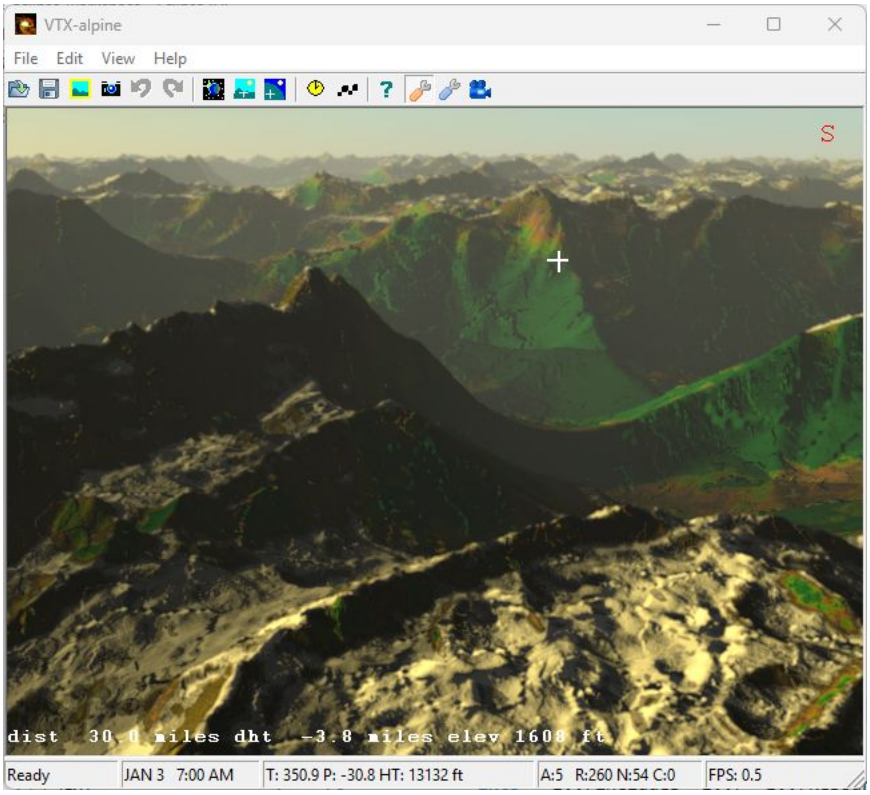
### Choosing where to go

Click the surface of a planet or moon, in orbit or surface view, and a white crosshair marks the point you clicked. A line of text at the bottom of the view shows the distance to the point (*dist*) and its elevation (*elev*).

Once a point is marked:

- **Surface** takes you to the crosshair and lowers you to the ground.
- **Selection** takes you to the crosshair but keeps your current height, so you see the spot from the same altitude.

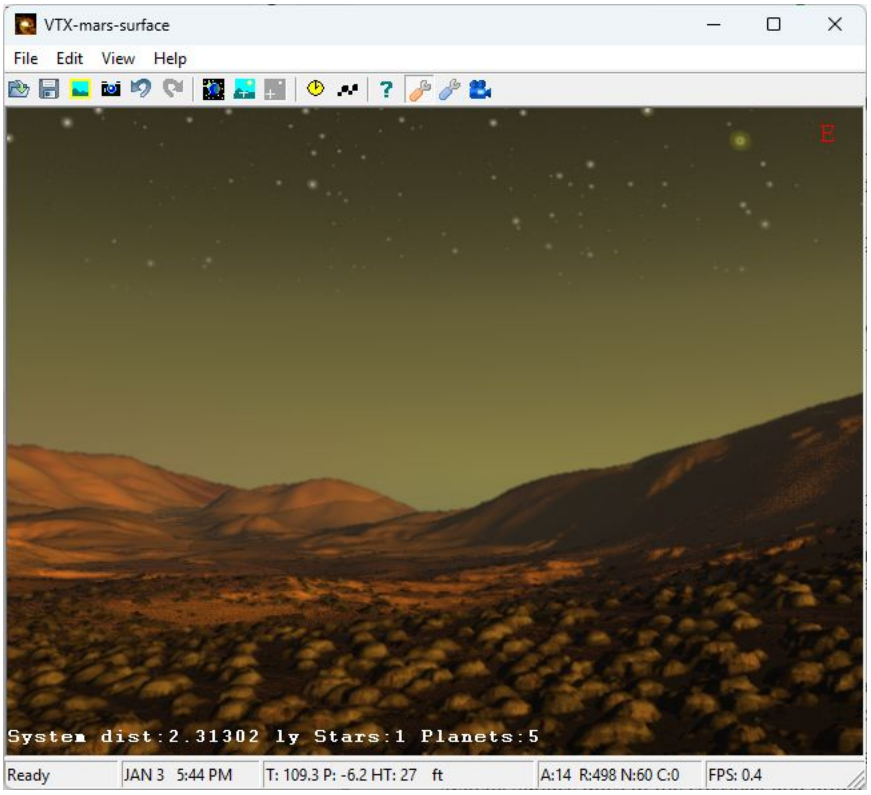
Both work the same way from orbit view and from surface view. Surface leaves you directly above the point, 1% of the distance away by default, so a point 30 miles off puts you about 0.3 miles above it. To approach more gently, lower **Approach** in the View box on the Scene tab of the Properties dialog: at 90, Surface covers 90% of the distance and leaves you 10% of it away. Pressing Surface again with no point marked halves your height. Clicking a planet or moon also selects it in the Properties dialog. Press **Space** to clear the selection.




*The crosshair marks a point on the surface; its distance and elevation show at the bottom*

## Traveling to the stars

Click a star in the star field and it turns yellow to show that it is selected. Only the nearer, foreground stars can be selected; the background stars can't. A line of text at the bottom of the view shows the distance to the star's system and how many stars and planets it has, for example *System dist:2.31302 ly Stars:1 Planets:5*.




*A selected star (yellow, upper right) and its system's distance, stars and planets at the bottom*

Then click **Orbit** (or press **Shift+U**). VTX transfers you to that system, with the system selected. When you arrive at a star that has never been visited, VTX generates its planetary system on the spot, so you can explore the universe one star at a time.

### *View history*

VTX remembers the views you visit, so you can return to them:

| Key | Action |
|---|---|
| = | Mark the current view |
| - / + | Go to the previous / next view in the history |
| _ | Rewind to the first view |
| z | Undo the last move |
| Z | Clear the view history |

**Edit > Undo** and **Edit > Redo** (and the toolbar buttons) step through the same history. **View > Clear History** clears it.

## Detail, quality and speed

VTX keeps refining what you see until the scene is as detailed as the current settings allow. A scene often

sharpens over a second or two after you stop moving.

### Scene quality

**View > Scene Quality** sets the overall level of detail:

| Setting | Key |
|---|---|
| Low | SHIFT-D |
| Medium | SHIFT-C |
| High | SHIFT-B |
| Very High | SHIFT-A |

**]** and **[** (**More Detail** and **Less Detail**) adjust the detail in smaller steps. Higher detail looks better but renders more slowly, so lower it while moving fast and raise it for pictures.

### Render options

The keyboard toggles rendering features on and off. The most useful are:

| Key | Feature |
|---|---|
| s | Shadows |
| f | Fog |
| x | Textures |
| b | Bump maps |
| l | Lighting |
| W | Water |
| a | Anti-aliasing |
| m | High dynamic range |
| S / N / L / P | Draw shaded (normal), solid, lines or points |
| R | Rebuild the scene |

With key help showing (**?**), the F1 list shows which features are on (**[X]**) and off (**[ ]**). The Scene object's tabs in the Properties dialog give finer control over the same settings.

**Rebuild** (**R**, or **Edit > Rebuild**) regenerates everything in view. Use it if the display looks wrong after many edits.

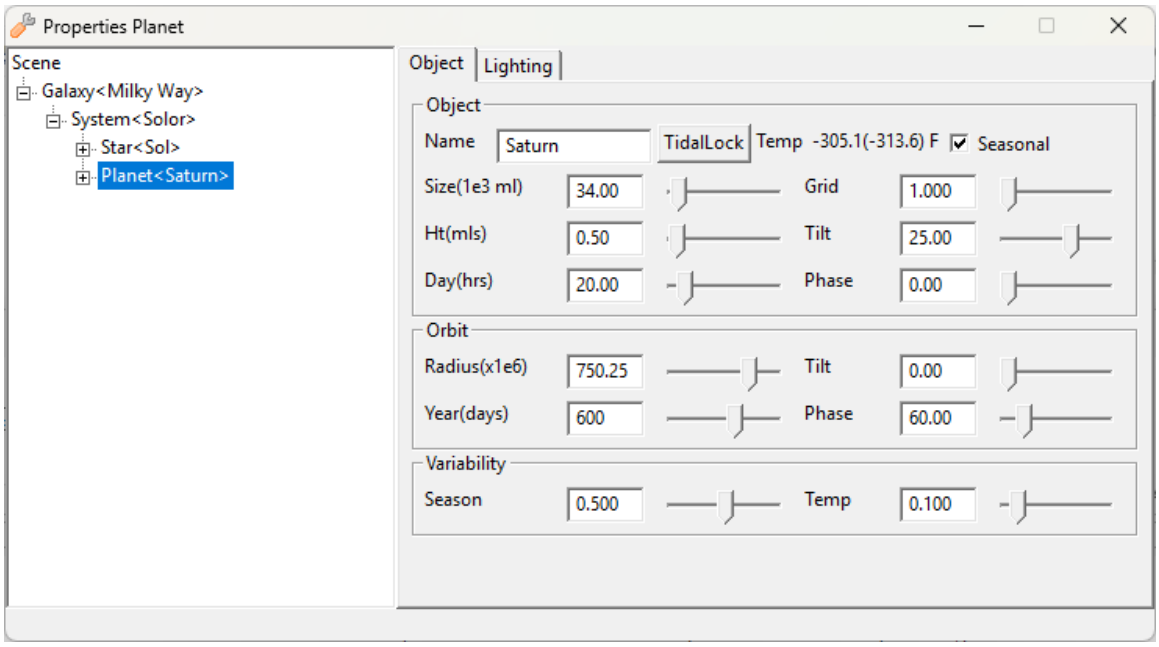
## Time and animation

Planets and moons rotate and orbit, and clouds drift. Time normally stands still, so you can study a view. To make time run:

- The **Time** toolbar button (or **View** > **Animate** > **Time**) starts and stops the clock.
- < and > decrease and increase the time rate. **:** resets it.
- The **Motion** toolbar button (or **View** > **Animate** > **Motion**) makes the view keep moving at the current speed, so you can glide over a surface hands-free.

The Scene object's **Display** tab shows the time and has a multiplier from x1 to x10^7 for fast-forwarding orbits.

# The Properties dialog




*The Properties dialog: the object tree on the left, a Planet's Object tab on the right*

The Properties dialog is where you inspect and change a scene. Open it with the **Properties** toolbar button or the **Tab** key.

## *The object tree*

The left side shows the scene as a tree, which mirrors how the universe nests: the Scene contains a Galaxy, which contains a System, which contains a Star and Planets, and a Planet contains its Moons, Sky, Clouds, Rings and a Surface with its layers and features.

Click an object in the tree to show its tabs on the right. Selecting an object in the view also selects it in the tree.

## *Changing values*

Each object type has its own set of tabs. Part II lists every tab and control. The controls work the same way everywhere:

- **Sliders** have a text box beside them. Drag the slider, or type a value in the box and press **Enter**.
- **Expression sliders** accept either a number or an expression, so a value can vary across a surface instead of being constant.
- **Color sliders** set a color with a color box and its intensity with a slider.
- **Expression fields** hold a complete terrain or color expression. Edit the text and press **Enter** to apply it.

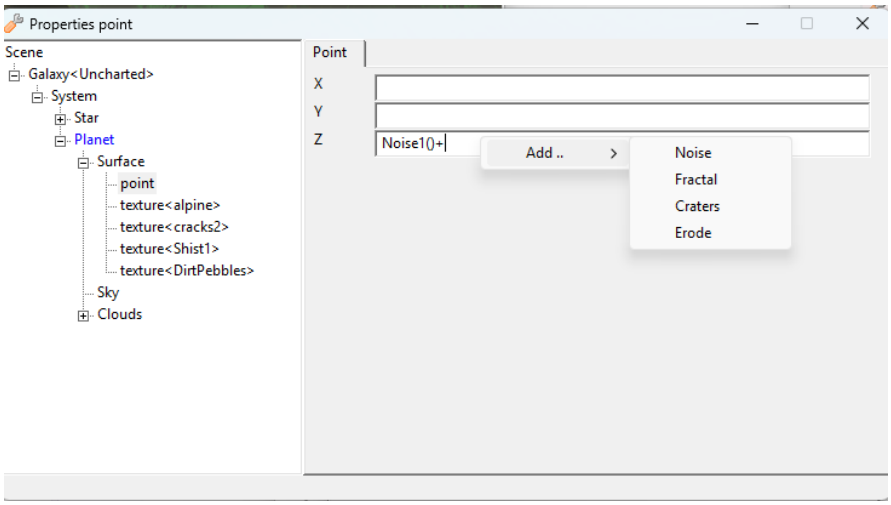
Most changes appear in the view immediately. Changes that affect terrain shape take a moment while the surface is rebuilt.

## Editing parts of an expression

Expressions in a surface can be long. VTX can show the noise, fractal, crater and erosion functions in an expression as short tokens such as *Noise1*, and edit each one in its own dialog:

1. Double-click a token in an expression field to select it.
2. Right-click and choose **Edit**. The Function Editor opens with that function's settings.
3. Change the settings. The expression updates as you change them.

To insert a new function, right-click in the field with nothing selected and choose **Add > Noise**, **Fractal**, **Craters** or **Erode**.



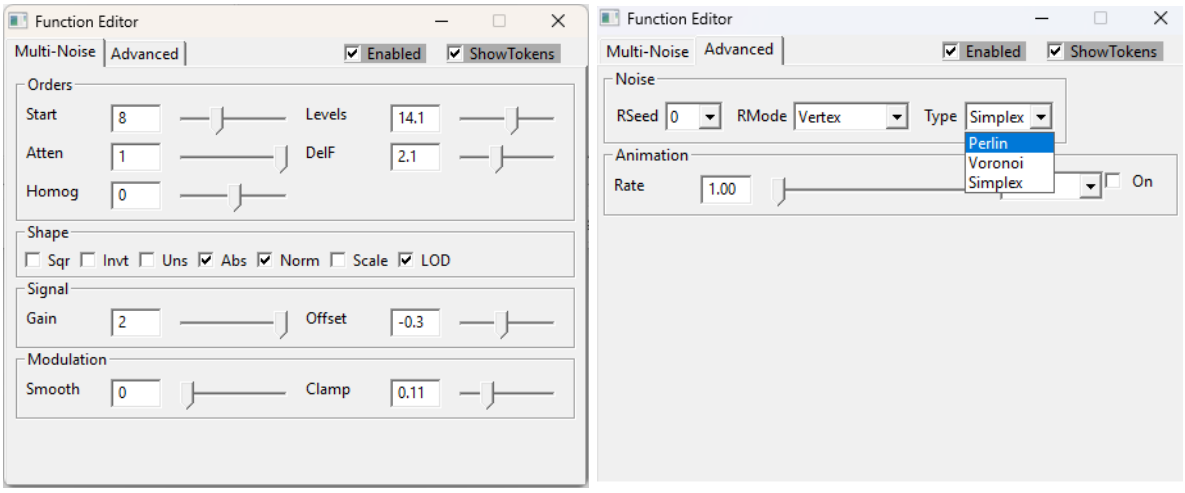

*Right-click an expression field and choose Add .. to insert a function (here the Z field of a surface point)*

The Function Editor has a tab with the function's main settings, such as **Multi-Noise** for a noise function, and an **Advanced** tab. For noise, the Advanced tab sets the noise type (Perlin, Voronoi or Simplex), the random seed and animation.

Two check boxes at the top work for every function. **Enabled** turns the function off without removing it from the expression. **ShowTokens** switches the expression fields between short tokens such as *Noise1* and

the function's full text.

For craters, the last line of the **Noise** group on the Craters tab gives access to the vertical and radial noise functions that roughen each crater. **Vertical noise..** and **Radial noise..** open that function in the noise editor, a second Function Editor window, where changes apply to the crater as you make them. **Default** puts back the built-in function, and is grayed out while the function is still the default. The Vertical and Radial sliders above set how strongly each function is applied.




*The Function Editor for a noise function: the Multi-Noise tab, and the Advanced tab with the noise types listed*

## The object menu

Right-click an object in the tree for its menu. What appears depends on the object type. The most common commands are:

| Command | Action |
|---|---|
| Set View | Make this object the center of the view |
| Show | Show or hide the object |
| Enable | Turn a feature (such as fog or erosion) on or off without deleting it |
| Delete | Remove the object from the scene |
| Add | Add a child object: a moon to a planet, a cloud layer, a plant, and so on |
| Replace with.. | Replace the object with a saved one or a newly generated one |
| Save.. | Save the object to the Objects library so you can reuse it |
| Randomize | Give the object new random settings of the same kind |
| Default | Reset the object to its default settings |
| Remove file.. | Delete a saved object from the Objects library |

You can also rearrange the tree by dragging objects with the mouse. Dragging with the right mouse button

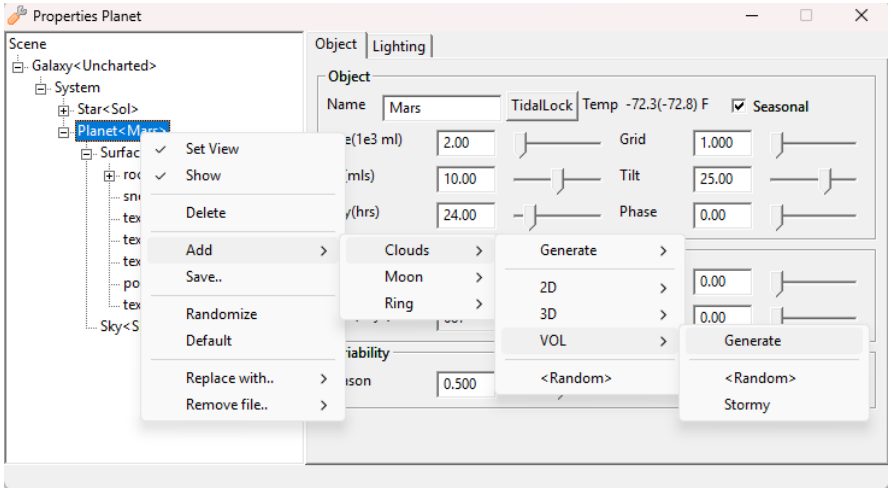
copies an object instead of moving it.

# Generating new worlds

Every star you travel to gets a generated planetary system. You can also generate objects deliberately:

- **Add > *type* > Generate** adds a new generated object, for example a new moon around a planet.
- **Replace with.. > Generate** replaces an object with a newly generated one of the same type. Some types offer sub-types, for example rocky, icy or gas giant planets.
- **Add** or **Replace with..** followed by **<Random>** picks one of your saved objects of that type at random.
- **Randomize** keeps the object's type but gives it new random settings.

If you don't like a result, generate again. Each generation is different.




*Adding a cloud layer to a planet: Add > Clouds > VOL > Generate. The same menu offers <Random> and saved clouds such as Stormy*

## *Reusing generated textures*

Generated worlds use texture images: height maps, erosion maps, layer images and color bands. Creating them takes time, so VTX can save them in the *Bitmaps* folder and reuse them for later worlds. Two settings on the Scene object's **Adapt** tab, in the *Procedural Generation* group, control this:

| Setting | Choices |
|---|---|
| Reuse | **All**: reuse saved images for everything. **2D** (default): reuse saved 2D maps, create new color bands. **None**: create every image new. |
| Keep | Which generated images to keep in *Bitmaps*: **All**, **2D** (default) or **None**. Images that aren't kept are deleted the next time VTX starts. |

With **2D**, reuse is phased in as *Bitmaps* fills up, so that a small collection doesn't make new worlds look alike. The first worlds always get new images. Once about ten images of a kind are saved, about half of new worlds reuse one, and with a hundred saved, about nine in ten do. Generation gets faster as the collection grows, and new images keep being added as long as Keep is on.

**All** reuses saved images whenever there is at least one of each kind, so use it only once *Bitmaps* has a good collection. Set Reuse to **None** if you want every new world to get new images.

The **Clear** button on the same line deletes the generated images in *Bitmaps* that the current scene doesn't use, so you can start a new collection. It keeps any image that a saved scene or object refers to without including its definition, since those would break. Saved scenes that do include an image's definition rebuild the image the next time they are opened, which takes a moment. Clear asks before deleting anything, and then reports how many images it deleted and how many it kept. Scenes and objects saved with this version of VTX include the definition of every generated image they use, including reused ones, so VTX can always rebuild their images.

# Scenes and saved objects

## *Opening and saving scenes*

A scene is everything in the universe you are exploring, plus the views you have saved. Scenes are stored as text files with the extension *.spx* in the *Saves* folder.

- **File > Open** opens a scene.
- **File > Save** saves the current scene.
- VTX saves the current session to *last.spx* when it quits, and reopens it the next time it starts.

Scene files are plain text that describes the object tree and its expressions, so they can also be read and edited in a text editor. The Theory of Operation describes the format.

## *Saved objects*

Any object can be saved on its own with **Save..** in its tree menu. Saved objects go into the *Objects* folder, organized by type, and then appear in the **Add** and **Replace with..** menus of matching objects. This is the way to build a library of favorite planets, cloud layers, plants or rock types and reuse them across scenes.

A new installation includes only the plants in *Objects\Plant*, which **Generate** draws on when it adds grass, bushes and trees to a world. The other folders are empty, so until you save objects of your own, the **Add** menus offer only the built-in default for those types. Scenes don't depend on the library, because a scene file contains all of its objects.

The VTX repository on GitHub (https://github.com/dwsindorf/VTX) has a collection of example objects in its *Objects* folder. To use them, copy the .spx files you want into the matching subfolder of your installation, for example *Objects\Planet*. Some were saved with earlier versions of VTX, so they may not load, or may look different from when they were made. Some also use textures that VTX generates into the *Bitmaps* folder, which a new installation doesn't have yet.

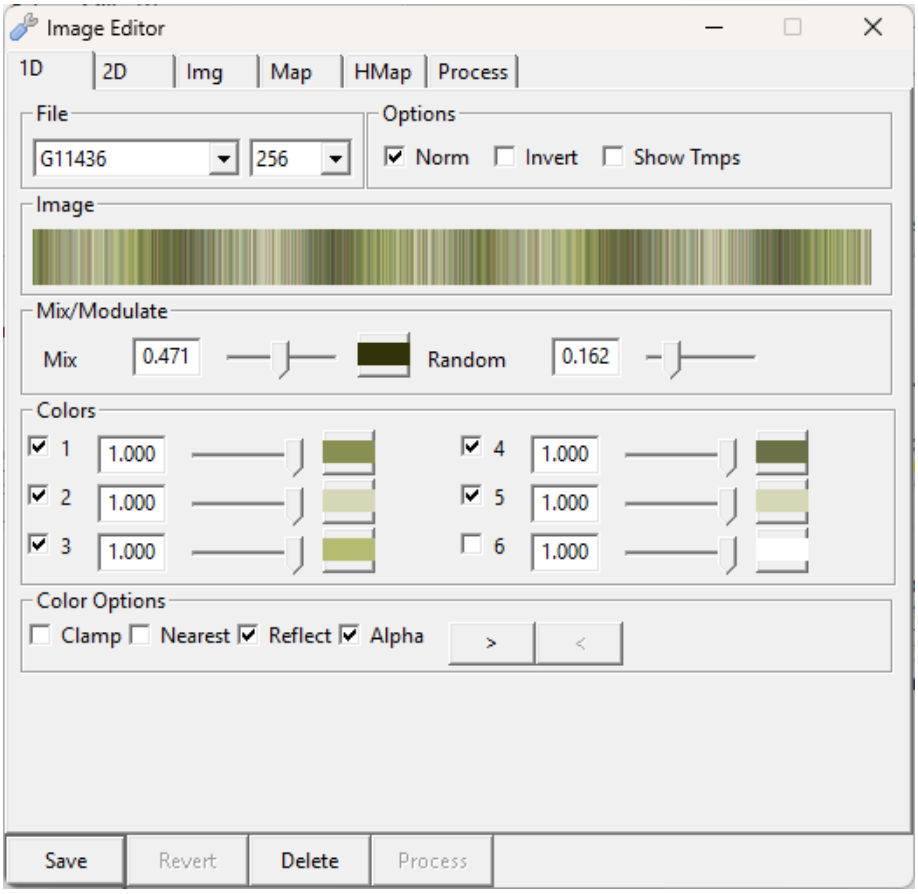
# Pictures and movies

## *Snapshots*

**File > Save Image** (the **Snapshot** toolbar button) saves the current view as an image in the *Images* folder. **File > View Image** opens a saved image.

For the best result, raise the scene quality and let the view settle before you save.

## *The Image Editor*



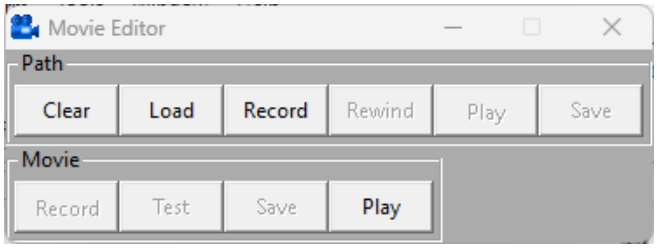

*The Image Editor, showing the 1D (bands) tab*

The Image Editor creates and edits the images that surfaces use as textures. Open it with the **Image Editor** toolbar button. Its tabs are:

| Tab | Purpose |
|---|---|
| 1D | Color bands: one-dimensional color gradients, used for layered and striped surfaces |
| 2D | Two-dimensional images generated from expressions |
| Img | Imported images from *Textures\Images* |

| Tab | Purpose |
|---|---|
| Map | Imported surface maps |
| HMap | Imported height maps |
| Process | Image processing (for example blurring) of images in *Textures\Processed* |

### *Movies*




*The Movie Editor*

VTX records a movie in two steps. First you record a *path* by flying through the scene. Then VTX replays the path and saves a frame at each step. Recording frames from a path gives smooth movement even when rendering each frame is slow.

Press **F3** to switch the number keys to Movie mode:

| Key | Action |
|---|---|
| 2 | Start recording a path. Move around the scene as you like. |
| 1 | Stop |
| 3 / 4 | Rewind / play the path |
| 5 / 6 | Save / load the path |
| 7 | Record movie frames from the path |
| 8 | Record movie frames live, as you move |
| 0 | Clear the path and movie |

The **Movie Editor** toolbar button opens the Movie Editor, which has the same commands as buttons: **Clear**, **Load**, **Record**, **Rewind**, **Play** and **Save** for paths, and **Record**, **Test**, **Save** and **Play** for movies. Frames are recorded as images in the *Movies* folder.

**Save** turns the recorded frames into a video file, and **Play** plays one back. Both use the free FFmpeg tools (*ffmpeg* and *ffplay*), which aren't included with VTX: install FFmpeg and make sure its *bin* folder is on your PATH. In the Play dialog, **Exit at end** closes the player when the video finishes.

# Troubleshooting

**The picture is coarse or slow.** Check that VTX is using the dedicated graphics card (see Graphics card

selection). Lower the scene quality while moving.

**Parts of the Properties dialog are cut off.** VTX's dialogs are laid out for 100% display scaling. If Windows display scaling (**Settings > System > Display > Scale**) is 125% or more, some controls may not fit. Setting the scale to 100% avoids this.

**VTX shows a shader error in its console.** The console lists the shader that failed. Make sure the *Shaders* folder is complete and writable. VTX writes a *.debug* copy of each shader there, which shows exactly what was sent to the graphics driver.

**Settings seem wrong after experimenting.** Press **0** in F1 mode to restore the display defaults, and **1** to restore the detail (adapt) defaults. Saved preferences are kept in *Resources\UI\prefs.spx*.

**The scene looks wrong after many edits.** Press **R** to rebuild it.

# Part II: Reference

This part is generated from the VTX source code (v1.0-3-g92b7cc9-dirty), so it matches this version of the program exactly. Ranges are the ranges of the sliders: you can often type values outside them.

## Menus and toolbar

### *File menu*

| Command | Key | Notes |
|---|---|---|
| Open .. | | |
| Save .. | | |
| Save Image .. | | |
| View Image .. | | |
| Exit | | |

### *Edit menu*

| Command | Key | Notes |
|---|---|---|
| Undo Z | | |
| Redo Y | | |
| Rebuild | SHIFT-R | |

### *View menu*

| Command | Key | Notes |
|---|---|---|
| Show Toolbar | | Show/Hide the toolbar |
| Clear History | | |
| Move to | | |
| Move to > Orbit | SHIFT-U | |
| Move to > Surface | SHIFT-G | |

| Command | Key | Notes |
|---|---|---|
| Move to > Selection | SHIFT-M | |
| Animate | | |
| Animate > Time | | |
| Animate > Motion | | |
| Window Size | | |
| Window Size > Small | | |
| Window Size > Medium | | |
| Window Size > Big | | |
| Window Size > Wide | | |
| Window Size > Panoramic | | |
| Scene Quality | | |
| Scene Quality > Low | SHIFT-D | |
| Scene Quality > Medium | SHIFT-C | |
| Scene Quality > High | SHIFT-B | |
| Scene Quality > Very High | SHIFT-A | |
| Scene Quality > More Detail | ] | |
| Scene Quality > Less Detail | [ | |

## Help menu

| Command | Key | Notes |
|---|---|---|
| key help ? | | Show/Hide Onscreen key help |
| About... | | Show about dialog |

## Toolbar

| Button | Action |
|---|---|
| Open | load a scene file (.spx) |
| Save | save the scene to a file (.spx) |
| View Image | show a saved JPEG or BMP image |
| Snapshot | save the current view as an image |
| Undo | go back to the previous view (-) |
| Redo | go forward to the next view (+) |
| Orbit | view the current object from orbit, or go to the selected star (Shift+U) |
| Surface | go down to the surface, at the crosshair if one is set (Shift+G) |
| Selection | go to the crosshair, keeping the current height (Shift+M) |

| Button | Action |
|---|---|
| Time | start or stop time animation |
| Motion | start or stop moving at the current speed |
| Key Help | show or hide the on-screen key list (?) |
| Properties | open the Properties dialog to edit the scene (Tab) |
| Image Editor | create and edit image textures |
| Movie Editor | record and play paths and movies |

# Keyboard

Press **?** to show the on-screen key help, and **F1** to **F6** to switch modes. The number keys **0** to **9** do different things in each mode. Letter keys are case-sensitive.

## *F1: All mode*

| Key | Action |
|---|---|
| 0 | Set display defaults |
| 1 | Set adapt defaults |
| ***Arrow keys*** | |
| left | Head left |
| right | Head right |
| up | Move forward |
| down | Move back |
| ***Standard keys*** | |
| u | Move up |
| d | Move down |
| [ | Decrease detail |
| ] | Increase detail |
| , | Decrease speed |
| . | Increase speed |
| < | Decrease time |
| > | Increase time (rate) |
| ; | Reset move rate |
| : | Reset time rate |
| R | Rebuild |
| W | Render water |
| P | Render points |

| Key | Action |
|---|---|
| L | Render lines |
| N | Render solid (ogl) S render solid (shaders) |
| ***Render options*** | |
| Render | Mode |
| P | Points L:lines N:triangles S:shaders |
| s | Shadows |
| f | Fog |
| b | Bumpmaps |
| x | Textures |
| n | Smoothing |
| l | Lighting |
| v | Ave normals |
| a | Antialiasing |
| ***Adapt options*** | |
| Scene | Detail |
| D | Draft C:average B:better A:best |
| o | Occlusion |
| e | LOD |

## *F2: View mode*

| Key | Action |
|---|---|
| 1 | Right turn |
| 2 | Left turn |
| 3 | Look left |
| 4 | Look right |
| 5 | Look down |
| 6 | Look up |
| 7 | Tilt left |
| 8 | Tilt right |
| 9 | Overhead view |
| 0 | Surface view (reset) |

## F3: Movie mode

| Key | Action |
|---|---|
| 0 | Clear (all) |
| 1 | Stop |
| ***Path*** | |
| 2 | Record |
| 3 | Rewind |
| 4 | Play |
| 5 | Save |
| 6 | Load |
| ***Movie*** | |
| 7 | Record (from path) |
| 8 | Record (live) |
| 9 | Test |

## F4: Stats mode

| Key | Action |
|---|---|
| 0 | Clear all |
| 1 | Objects |
| 2 | Maps |
| 3 | Cells |
| 4 | Raster |
| 5 | Stars |
| 6 | Placements |
| 7 | Terrain |
| 8 | Plants |

## F5: Colors mode

| Key | Action |
|---|---|
| 0 | Normal colors |
| 1 | Node size |
| 2 | Node type |
| 3 | Node visibility |
| 4 | Node layers |
| 5 | Node edges |

| Key | Action |
|---|---|
| 6 | Node margins |
| 7 | Node tests |

## F6: Test mode

| Key | Action |
|---|---|
| 0 | Normal view |
| 1 | Laboratory view |
| 2 | Light source view |

## All key commands

Every single-key command, as defined in the program. Commands marked *developer* are for testing and diagnostics.

| Key | Command |
|---|---|
| F1 | Key help mode: All (default) |
| F2 | Key help mode: View |
| F3 | Key help mode: Movie |
| F4 | Key help mode: Stats |
| F5 | Key help mode: Colors |
| F6 | Key help mode: Test |
| F7 | F7 (*developer*) |
| F8 | F8 (*developer*) |
| F9 | F9 (*developer*) |
| F10 | F10 (*developer*) |
| Left arrow | Turn (head) left |
| Right arrow | Turn (head) right |
| Up arrow | Move forward |
| Down arrow | Move back |
| ? | Show or hide the on-screen key help |
| ! | Test1 (*developer*) |
| @ | Test2 (*developer*) |
| # | Test3 (*developer*) |
| $ | Test4 (*developer*) |
| % | Test5 (*developer*) |
| ^ | Test6 (*developer*) |

| Key | Command |
|---|---|
| & | Test7 (*developer*) |
| * | Test8 (*developer*) |
| Q | Make the default scene (at startup) |
| u | Move up |
| d | Move down |
| Ctrl+u | Tilt the view up (also Ctrl+Up) |
| Ctrl+d | Tilt the view down (also Ctrl+Down) |
| M | Move to the selected object |
| _ | Rewind to the first saved view |
| - | Previous view in the history |
| + | Next view in the history |
| Space | Clear the selection |
| = | Mark (save) the current view |
| z | Undo the last move |
| Z | Clear the view history |
| I | New random seed |
| y | Yydebug (*developer*) |
| q | Quit |
| . | Increase movement speed |
| , | Decrease movement speed |
| > | Increase the time rate |
| < | Decrease the time rate |
| g | Start or stop continuous motion |
| t | Start or stop time |
| ; | Reset movement speed |
| : | Reset the time rate |
| ] | More detail |
| [ | Less detail |
| G | Surface view |
| O | Orbit view |
| U | Orbit (global) view |
| V | Vertical view |
| H | Horizontal view |
| \ | Raster shadow map (*developer*) |

| Key | Command |
|---|---|
| k | Raster lines (*developer*) |
| S | Draw shaded (shaders) |
| N | Draw solid (OpenGL) |
| L | Draw lines |
| P | Draw points |
| W | Water on/off |
| w | Water depth on/off |
| c | Mix (*developer*) |
| x | Textures on/off |
| X | Sprites on/off |
| Y | Specular light on/off |
| b | Bump maps on/off |
| r | Water reflections on/off |
| l | Lighting on/off |
| n | Smooth shading on/off |
| s | Shadows on/off |
| T | Twilight on/off |
| f | Fog on/off |
| v | Averaged normals on/off |
| i | Nvis (*developer*) |
| F | Face (*developer*) |
| a | Anti-aliasing on/off |
| m | High dynamic range on/off |
| j | Geometry shader terrain on/off |
| o | Occlusion test on/off |
| e | Level of detail on/off |
| E | Uniform (*developer*) |
| h | Hidden (*developer*) |
| R | Rebuild the scene |
| A | Scene quality: Very High |
| B | Scene quality: High |
| C | Scene quality: Medium |
| D | Scene quality: Low |

# Properties dialog

Each object type in the scene tree has its own tabs. For each type, this chapter lists its right-click menu, then every tab, grouped as in the dialog.

## *Scene*

**Menu:** Add

### Display tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Animation*** | | | |
| Time | slider | 1 to 10 | |
| Animation | list | x1, x10, x100, x1000, x10^4, x10^5, x10^6, x10^7 | |
| Animation | list | x1, x10, x100, x1000, x10^4, x10^5, x10^6, x10^7 | |
| Rate | slider | 1 to 10 | |
| + | checkbox | | |
| ***View*** | | | |
| FOV | slider | 20 to 75 | |
| Approach | slider | 10 to 99 | |
| Draw Mode | option buttons | Points, lines, Solid, Shaded | |
| Lighting Model | option buttons | OGL, Phong, Mix | |

### Adapt tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Vertex Generation*** | | | |
| Detail | list | Draft, Normal, High, Photo | |
| CellSize | slider | 1 to 10 | |
| Geometry Level | list | 1, 2, 3, 4, 5, 6 | |
| ***Procedural Generation*** | | | |
| Detail | list | Low, Medium, High, Max | |
| Layers | list | Auto, 1, 2, 3, 4 | |
| Reuse | list | All, 2D, None | Which saved generated images new worlds reuse (see Generating new worlds). |
| Keep | list | All, 2D, None | Which generated images are kept in the Bitmaps folder. |
| Clear | button | | Delete generated images the current scene doesn't use |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Adapt Tests*** | | | |
| Occlusion | checkbox | | Decrease resolution in occluded areas |
| Clipping | checkbox | | Decrease resolution in off-screen areas |
| Backfacing | checkbox | | Decrease resolution in backfacing sides |
| Curvature | checkbox | | Decrease resolution in smooth areas |
| Adapt Tests | option buttons | Overlap, Merge | |

## Render tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Render Options*** | | | |
| HDR | checkbox | | Enable High dynamic range |
| Shadows | checkbox | | Enable shadows |
| Water | checkbox | | Enable water |
| ***High Dynamic Range*** | | | |
| Min | slider | 0.0 to 30.0 | |
| Max | slider | 0.0 to 10.0 | |
| ***Shadows*** | | | |
| Blur | slider | 0.1 to 2.0 | |
| Views | list | 1, 2, 3, 4, 5, 6, 7, 8 | |
| Cascades | checkbox | | Use cascades |
| FOV | slider | 0.5 to 1.5 | |
| DOV | slider | 0.1 to 1.5 | |

## Filter tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Filter Options*** | | | |
| AveNorms | checkbox | | Average normals at each vertex |
| Aniso | checkbox | | Anisotropic texture filtering |
| Edges | checkbox | | Apply image filter |
| Big Kernel | checkbox | | use large filter kernel |
| Show Mask | checkbox | | Show image filter |
| ***Edges*** | | | |
| Color | slider | 0.0 to 1.0 | |
| Normal | slider | 0.0 to 1.0 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Textures*** | | | |
| MIP | slider | -1 to 1.0 | |
| Freq | slider | -3 to 1.0 | |
| Color | slider | -1 to 1.0 | |
| Bump | slider | -1.0 to 1.0 | |

### Options tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Grid Options*** | | | |
| Map | checkbox | | Draw Long/Lat lines |
| Contours | checkbox | | Draw Contour lines |
| Auto | checkbox | | Autoset intervals |
| Grid Options | option buttons | K, C, F | |
| ***Lines*** | | | |
| Grid | slider | 0.1 to 10 | |
| Miles | color box | | |
| Lat | color box | | |
| Contours | slider | 100 to 20000 | |
| Feet | color box | | |
| Text | color box | | |

## *Planet*

**Menu:** Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| TidalLock | button | | Make the year (orbit) as long as the day and reset the rotation phase, so one side always faces the star. |
| Temp | display only | | |
| Seasonal | checkbox | | |
| Size(1e3 ml) | slider | 1 to 1000 | Planet size in thousands of miles. |
| Grid | slider | 1 to 4 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Ht(mls) | slider | 0.1 to 20 | Maximum terrain height in miles. |
| Tilt | slider | -90 to 90 | Tilt of the rotation axis in degrees; gives the planet seasons. |
| Day(hrs) | slider | 0 to 200 | Length of the planet's day (rotation period) in hours. |
| Phase | slider | 0 to 360 | |
| ***Orbit*** | | | |
| Radius(x1e6) | slider | 1 to 1000 | Orbit radius in millions of miles. |
| Tilt | slider | 0 to 90 | Inclination of the orbit in degrees. |
| Year(days) | slider | 0 to 1000 | Length of the orbit in days. |
| Phase | slider | 0 to 360 | |
| ***Variability*** | | | |
| Season | slider | 0.0 to 1 | |
| Temp | slider | 0.0 to 1 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Color*** | | | |
| Shine | slider | 0.25 to 100 | |
| Albedo | slider | 0.01 to 1 | |
| Ambient | color slider | | |
| Specular | color slider | | |
| Diffuse | color slider | | |
| Radiance | color slider | | |
| Shadow | color slider | | |

## *Asteroid*

**Menu:** Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| MaxDepth | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Smooth | checkbox | | Smooth normals |
| Lines | checkbox | | wireframe |
| Size(miles) | slider | 1 to 1000 | |
| Grid | slider | 1 to 4 | |
| Day(hrs) | slider | 0 to 200 | |
| Phase | slider | 0 to 360 | |
| ***Shape*** | | | |
| Scale | slider | 0.0 to 1 | |
| RNoise | expression field | | |
| Symmetry | slider | 0.2 to 1 | |
| VNoise | expression field | | |
| ***Orbit*** | | | |
| Radius(x1e6) | slider | 1 to 1000 | |
| Tilt | slider | 0 to 90 | |
| Year(days) | slider | 0 to 1000 | |
| Phase | slider | 0 to 360 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Color*** | | | |
| Shine | slider | 0.25 to 100 | |
| Ambient | color slider | | |
| Specular | color slider | | |
| Diffuse | color slider | | |
| Shadow | color slider | | |

## *Star*

**Menu:** Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| Type | display only | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Size(mm) | slider | 0.1 to 10 | |
| Grid | slider | 1 to 4 | |
| Day(hrs) | slider | 0 to 200 | |
| Phase | slider | 0 to 360 | |
| ***Orbit*** | | | |
| Radius(mm) | slider | 0 to 100 | |
| Oblique | slider | 0 to 0.95 | |
| Year(days) | slider | 0 to 1000 | |
| Phase | slider | 0 to 360 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Color*** | | | |
| Radiance | color slider | | |
| Specular | color slider | | |
| Diffuse | color slider | | |

## *Moon*

**Menu:** Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| TidalLock | button | | |
| Temp | display only | | |
| Seasonal | checkbox | | |
| Size(mls) | slider | 100 to 10000 | |
| Grid | slider | 1 to 4 | |
| Ht(mls) | slider | 0.1 to 20 | |
| Tilt | slider | -90 to 90 | |
| Day(hrs) | slider | 0 to 200 | |
| Phase | slider | 0 to 360 | |
| ***Orbit*** | | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Radius(x1e3) | slider | 1 to 100 | |
| Tilt | slider | 0 to 90 | |
| Year(days) | slider | 0 to 1000 | |
| Phase | slider | 0 to 360 | |
| ***Variability*** | | | |
| Season | slider | 0.0 to 1 | |
| Temp | slider | 0.0 to 1 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Color*** | | | |
| Shine | slider | 0.25 to 100 | |
| Albedo | slider | 0.01 to 1 | |
| Ambient | color slider | | |
| Specular | color slider | | |
| Diffuse | color slider | | |
| Radiance | color slider | | |
| Shadow | color slider | | |

## *Corona*

**Menu:** Show, Delete

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Geometry*** | | | |
| Name | text | | |
| Radius | slider | 1 to 10 | |
| ***Color*** | | | |
| Inner | color slider | | |
| Outer | color slider | | |
| ***Gradient*** | | | |
| Alpha | slider | 0.0 to 1 | |
| Internal | checkbox | | |
| ***Density*** | | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Density | expression field | | |

## *Halo*

**Menu:** Show, Delete

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Geometry*** | | | |
| Name | text | | |
| Ht | slider | 0.001 to 0.1 | |
| ***Color*** | | | |
| Inner | color slider | | |
| Outer | color slider | | |
| ***Gradient*** | | | |
| Alpha | slider | 0.0 to 1 | |

## *Clouds*

**Menu:** Show, Delete, Save.., Add, Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Orbital*** | | | |
| Name | text | | |
| Orbital | list | 2D, 3D, Vol | 2D: textured layer, 3D: sprites, Vol: ray-marched volume |
| DP | checkbox | | Double precision noise: smooth noise in close views (2D, Vol), but much slower for Vol |
| Grid | slider | 1 to 8 | |
| Height | slider | 10 to 100 | |
| Tilt | slider | 0 to 360 | |
| Period | slider | -20 to 20 | |
| Phase | slider | 0 to 360 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Color*** | | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Shine | slider | 0.25 to 100 | |
| Ambient | color slider | | |
| Specular | color slider | | |
| Diffuse | color slider | | |
| Radiance | color slider | | |

## Shape (shown only for some modes) tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***3D Shape*** | | | |
| 3D Shape | expression field | | |
| Top | expression field | | |
| ***Sprite Properties*** | | | |
| Number | slider | 1 to 8 | |
| Scatter | slider | 0 to 32 | |
| Size | slider | 1 to 128 | |
| Range | slider | 1 to 4 | |
| Angle | slider | 0 to 1 | |
| Darken | slider | 0 to 2 | |

## Images (shown only for some modes) tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Sprites*** | | | |
| File | list | | |
| Sprites | list | 1x, 4x, 16x | |
| Type expr | expression field | | |

## Volume (shown only for some modes) tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Height*** | | | |
| Ampl | slider | 0.5 to 15 | |
| Var | slider | 0 to 15 | |
| ***Shape*** | | | |
| Freq | slider | 0.1 to 3 | |
| Erosion | slider | 0 to 1 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Density | slider | 0.25 to 2 | |
| Darken | slider | 0 to 2 | |
| Steps | slider | 16 to 256 | |
| Cover | slider | -1 to 1 | |

## *Sky*

**Menu:** Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| Grid | slider | 1 to 4 | |
| Height | slider | 10 to 200 | |
| Opacity | slider | 0 to 1 | |
| Pressure | slider | 0 to 200 | |
| % GHG | slider | 0 to 100 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Color*** | | | |
| Halo | color slider | | |
| Sky | color slider | | |
| Twilite | color slider | | |
| Night | color slider | | |
| Scatter | color slider | | |
| Sunset | color slider | 0 to 4 | |
| ***Sharpen*** | | | |
| Sharpen | slider | 0.1 to 5 | |
| Sky grad | slider | 0.01 to 1 | |
| Twilite grad | slider | 0.01 to 1 | |
| Night grad | slider | -0.01 to -1 | |
| Scatter grad | slider | 0 to 1 | |
| Redden grad | slider | 0.2 to 5 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Haze*** | | | |
| Color | color slider | | |
| Haze | slider | 0 to 1 | |
| Min | slider | 0 to 5000 | |
| Depth | slider | 0 to 100000 | |

## *Texture*

**Menu:** Enable, Delete, Save.., Add

### Image tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Image*** | | | |
| Image | list | | |
| Image | list | | |
| Seasonal | checkbox | | |
| ***Color*** | | | |
| Ampl | slider | 0 to 2.0 | |
| Offset | slider | -2.0 to 2.0 | |
| Color | checkbox | | |
| ***Bump*** | | | |
| Ampl | slider | -8 to 8 | |
| Offset | slider | 0 to 1 | |
| Bump | checkbox | | |
| ***HtMap*** | | | |
| Ampl | slider | -50 to 50 | |
| Offset | slider | -1 to 1 | |
| HtMap | checkbox | | |
| ***Overlays*** | | | |
| Start | slider | -1 to 29 | |
| Levels | slider | 1 to 25 | |
| Ampl | slider | 0.0 to 1.0 | |
| Freq | slider | 1.0 to 5.0 | |

### Mapping tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Interpolation | option buttons | Nearest, Linear, Mip | |
| ***Options*** | | | |
| Clamp | checkbox | | |
| Normalize | checkbox | | |
| Dealias | checkbox | | |
| Triplanar | checkbox | | |
| ***Modulation*** | | | |
| Modulation | expression field | | |
| Alpha expr | expression field | | |
| ***Env. Bias*** | | | |
| Lat | slider | -1 to 1 | |
| Height | slider | -1 to 1 | |
| Bump | slider | -1 to 1 | |
| Slope | slider | -1 to 1 | |

## *Sprite*

**Menu:** Enable, Delete, Save.., Add

### Distribution tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Multilevel*** | | | |
| Levels | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | |
| Size | slider | 1 to 200 | |
| Density | expression slider | 0.0 to 1.0 | |
| ***Dispersion*** | | | |
| Size | expression slider | 0.0 to 2 | |
| Drop | expression slider | 0 to 2 | |
| ***Density Bias*** | | | |
| Slope | expression slider | -1 to 1 | |
| Hardness | expression slider | -1 to 1 | |
| Latitude | expression slider | -1 to 1 | |
| Height | expression slider | -1 to 1 | |

### Image tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Sprites*** | | | |
| File | list | | |
| Sprites | list | | |
| Center | list | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 | |
| Bias | expression slider | 0 to 1 | |

## *Plant*

**Menu:** Enable, Delete, Save.., Add, Randomize, Default, Replace with.., Remove file..

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Properties*** | | | |
| Name | text | | |
| Levels | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | |
| Density | expression slider | 0.0 to 1.0 | |
| ***Size*** | | | |
| Max | slider | 1 to 200 | |
| Delta | expression slider | 0.0 to 2 | |
| ***Density Bias*** | | | |
| Slope | expression slider | -1 to 1 | |
| Hardness | expression slider | -1 to 1 | |
| Latitude | expression slider | -1 to 1 | |
| Height | expression slider | -1 to 1 | |
| ***Other*** | | | |
| Drop | expression slider | 0 to 2 | |
| Skeleton | checkbox | | |
| Lines | checkbox | | |
| Splines | checkbox | | |
| One | checkbox | | |

## *Branch*

**Menu:** Enable, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

## Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Name | text | | |
| Randomize | checkbox | | |
| ***Levels*** | | | |
| Max | list | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 | |
| Min | list | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 | |
| End | checkbox | | |
| First | list | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 | |
| x10 | checkbox | | |
| Later | list | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 | |
| ***Size*** | | | |
| Length | expression slider | 0.1 to 2 | |
| Taper | expression slider | 0.1 to 1 | |
| Width | expression slider | 0.1 to 1 | |
| Taper | expression slider | 0.1 to 1 | |
| ***Dispersion*** | | | |
| Divergence | expression slider | 0 to 1 | |
| Offset | expression slider | 0 to 1 | |
| Random | expression slider | 0 to 1 | |
| Density | expression slider | 0 to 1 | |
| ***Curvature*** | | | |
| Amplitude | expression slider | 0 to 1 | |
| Direction | expression slider | -1 to 1 | |

## Texture tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Image*** | | | |
| File | list | | |
| Image | list | | |
| Enable | checkbox | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Center | expression slider | 0 to 1 | |

### Color tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Red | expression field | | |
| Green | expression field | | |
| Blue | expression field | | |
| Alpha | expression field | | |
| Color picker | color box | | |
| Revert | button | | |
| Enable | checkbox | | |

## *Leaf*

**Menu:** Enable, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Name | text | | |
| Cluster | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | |
| X | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | |
| Shape | checkbox | | |
| Randomize | checkbox | | |
| ***Shape*** | | | |
| Length | expression slider | 1 to 50 | |
| Taper | expression slider | 0.0 to 1 | |
| Width | expression slider | 0.2 to 5 | |
| Taper | expression slider | 0.1 to 1 | |
| ***Dispersion*** | | | |
| Divergence | expression slider | 0 to 1 | |
| Offset | expression slider | 0 to 1 | |
| Random | expression slider | 0 to 1 | |
| Density | expression slider | 0 to 1 | |
| ***Curvature*** | | | |
| Amplitude | expression slider | 0 to 1 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Direction | expression slider | -1 to 1 | |

### Texture tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Image*** | | | |
| File | list | | |
| Image | list | | |
| Enable | checkbox | | |
| Center | expression slider | 0 to 1 | |

### Color tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Red | expression field | | |
| Green | expression field | | |
| Blue | expression field | | |
| Alpha | expression field | | |
| Color picker | color box | | |
| Revert | button | | |
| Enable | checkbox | | |

## *Color*

**Menu:** Enable, Delete

### Color tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Red | expression field | | |
| Green | expression field | | |
| Blue | expression field | | |
| Alpha | expression field | | |
| Auto check | checkbox | | |
| Auto | color box | | |
| Revert | button | | |

## *Point*

**Menu:** Enable, Delete

### Point tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| X | expression field | | |
| Y | expression field | | |
| Z | expression field | | |

## *Gloss*

**Menu:** Enable, Delete

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Shine | slider | 0.5 to 20 | |
| Albedo | slider | 0 to 1 | |

## *Hardness*

**Menu:** Enable, Delete

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Density | slider | 0.0 to 1 | |

## *Layer*

**Menu:** Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Layer tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Properties*** | | | |
| Label | text | | |
| Square transition slope | checkbox | | |
| ***Width*** | | | |
| Width | expression slider | 0 to 1.0 | |
| ***Morph*** | | | |
| Morph | expression slider | 0 to 1.0 | |
| ***Drop*** | | | |
| Drop | expression slider | 0 to 1.0 | |
| ***Ramp*** | | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Ramp | expression slider | 0 to 1.0 | |

## Map

**Menu:** Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Transfer Functions*** | | | |
| Height | expression slider | 0 to 1.0 | |
| Base | expression slider | -1.0 to 1.0 | |
| Scale | expression slider | 0.0 to 1.0 | |
| Margin | expression slider | 0.0 to 4.0 | |
| Layer Intersection Mode | option buttons | Overlap, Merge | |

## Fog

**Menu:** Enable, Delete, Save.., Randomize, Default, Replace with.., Remove file..

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Horizontal Range (feet)*** | | | |
| Min | slider | 0 to 1000 | |
| Depth | slider | 0 to 10000 | |
| ***Vertical Range (feet)*** | | | |
| Min | slider | -5000 to 5000 | |
| Depth | slider | 0 to 10000 | |
| ***Properties*** | | | |
| Density | expression field | | |
| Color | color slider | | |
| Glow | slider | 0 to 1 | |

## Water

**Menu:** Enable, Delete, Save.., Randomize, Default, Replace with.., Remove file..

## Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| Level | slider | -10000 to 10000 | |
| Object | option buttons | Gas, Liquid, Solid | |
| Auto | checkbox | | |
| Planet temp | display only | | |
| Sky View Reflections | checkbox | | Render an extra view above the screen so reflections of off-screen sky and terrain don't fall back to an estimated sky color |
| ***Presets*** | | | |
| Presets | list | | |
| Modulation | expression field | | |
| Default | button | | |
| ***Volatility*** | | | |
| Evaporation | slider | 0.1 to 10 | |
| Sublimation | slider | 0.1 to 10.0 | |

## Solid tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Surface*** | | | |
| Modulation | expression field | | |
| Default | button | | |
| Mix | slider | 0 to 1.0 | |
| ***Transparency*** | | | |
| feet | color slider | 1 to 1000 | |
| ***Reflection*** | | | |
| % | color slider | 0 to 1 | |
| ***Specular*** | | | |
| shine | slider | 0 to 100.0 | |
| albedo | slider | 0 to 1.0 | |
| ***Temperature (C)*** | | | |
| Liquid | slider | -1000 to 1000 | |
| Trans | slider | 0 to 10.0 | |

### Liquid tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Surface*** | | | |
| Modulation | expression field | | |
| Default | button | | |
| Mix | slider | 0 to 1.0 | |
| ***Transparency*** | | | |
| feet | color slider | 1 to 1000 | |
| ***Reflection*** | | | |
| % | color slider | 0 to 1 | |
| ***Specular*** | | | |
| shine | slider | 0 to 100.0 | |
| albedo | slider | 0 to 1.0 | |
| ***Temperature (C)*** | | | |
| Gas | slider | -1000 to 1000 | |
| Trans | slider | 0 to 10.0 | |

## *Surface*

**Menu:** Add, Save.., Randomize, Default, Replace with.., Remove file..

### Controls tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Variables*** | | | |
| Variables | list | | |
| Expr | expression field | | |
| New | button | | |
| Delete | button | | |
| Save | button | | |

## *Rocks*

**Menu:** Enable, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Rocks tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Properties*** | | | |
| Name | text | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Levels | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | |
| Density | expression slider | 0.0 to 1.0 | |
| ***Size*** | | | |
| Max | slider | 0.1 to 0.9999 | |
| Size | list | 10, 1, 0.1, 0.01, 0.001, 1e-4, 1e-5, 1e-6, 1e-7 | |
| Delta | expression slider | 0.0 to 2.0 | |
| ***Shape*** | | | |
| Ampl | expression slider | 0.0 to 1.0 | |
| Noise | expression field | | |
| Smooth | checkbox | | Smooth normals |
| Compress | expression slider | 0 to 1.0 | |
| Drop | expression slider | 0 to 1 | |
| 3D | checkbox | | |
| ***Density Bias*** | | | |
| Slope | expression slider | -1 to 1 | |
| Hardness | expression slider | -1 to 1 | |
| Latitude | expression slider | -1 to 1 | |
| Height | expression slider | -1 to 1 | |

## Rocks 3D

Uses the same tabs as **Rocks**.

## Ring

**Menu:** Show, Delete, Save.., Add

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Object*** | | | |
| Name | text | | |
| Grid | slider | 1 to 4 | |
| Radius | slider | 0.1 to 2 | |
| Width | slider | 0.1 to 2 | |

### Lighting tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Lighting*** | | | |
| Emission | color slider | | |
| Shadow | color slider | | |

## *Snow*

**Menu:** Enable, Delete

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Snow*** | | | |
| Level | slider | -4 to 4.0 | |
| Depth | slider | 0.0 to 1.0 | |
| ***Bias*** | | | |
| Lat | slider | 0.0 to 1.0 | |
| Slope | slider | -1 to 1 | |
| Ht | slider | -4 to 4.0 | |
| Bump | slider | -1 to 1.0 | |

## *System*

**Menu:** Set View, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Properties tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Name | text | | |
| ***Origin*** | | | |
| X | slider | -10000.0 to 10000.0 | |
| Y | slider | -100.0 to 100.0 | |
| Z | slider | -1000.0 to 1000.0 | |

### View tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Set View*** | | | |
| Top | button | | |
| Edge | button | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Oblique | button | | |

## *Galaxy*

**Menu:** Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

### Object tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Name | text | | |
| Type | display only | | |
| Size(ly) | slider | 100 to 100000 | |
| Grid | slider | 0.25 to 4 | |
| ***Origin*** | | | |
| X | slider | -100000.0 to 100000.0 | |
| Y | slider | -100000.0 to 100000.0 | |
| Z | slider | -100000.0 to 100000.0 | |
| ***Rotation*** | | | |
| X | slider | 0 to 360 | |
| Y | slider | 0 to 360 | |
| Z | slider | 0 to 360 | |

### View tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Set View*** | | | |
| Core | button | | |
| Rim | button | | |
| Top | button | | |
| Edge | button | | |
| Oblique | button | | |
| ***Position Radius-Theta-Phi*** | | | |
| R | slider | 0 to 100000.0 | |
| T | slider | 0 to 360 | |
| P | slider | 0 to 360 | |
| ***Rotation X-Y-Z*** | | | |
| X | slider | 0 to 360 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Y | slider | 0 to 360 | |
| Z | slider | 0 to 360 | |

## Shape tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Default Shape*** | | | |
| Spiral | button | | |
| Barred | button | | |
| Globular | button | | |
| Elyptical | button | | |
| Irregular | button | | |
| ***Radius*** | | | |
| Core | slider | 0 to 1 | |
| Rim | slider | 0 to 1 | |
| ***Twist*** | | | |
| Radius | slider | 0 to 2.0 | |
| Angle | slider | 0 to 720 | |
| ***Surface*** | | | |
| In | slider | 0 to 1 | |
| Out | slider | 0.0001 to 0.1 | |
| ***Compression*** | | | |
| Vertical | slider | 0 to 1.0 | |
| Radial | slider | 0 to 1 | |

## Density tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Gradient*** | | | |
| Vertical | slider | 0 to 2 | |
| Radial | slider | 0 to 2 | |
| ***Distribution*** | | | |
| Diffusion | slider | 0.01 to 4 | |
| Scatter | slider | 0 to 20 | |
| ***Noise*** | | | |
| Amplitude | slider | 0 to 1 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Vertical | slider | 0 to 1 | |
| Saturation | slider | 0 to 4 | |
| Cutoff | slider | -1 to 1 | |
| Noise | expression field | | |
| Default | button | | |

### Render tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Star Size*** | | | |
| Near | slider | 0 to 5 | |
| Far | slider | 0 to 5 | |
| ***Nova*** | | | |
| Size | slider | 0 to 5 | |
| % | slider | 0 to 1 | |
| ***Stars*** | | | |
| Near | slider | 10 to 500 | |
| Size | slider | 0.5 to 4 | |
| Far | slider | 1 to 5 | |
| Dust | slider | 0.0 to 4 | |
| ***Color*** | | | |
| Color | expression field | | |
| Intensity | slider | 0 to 1 | |
| Bias | slider | 0 to 2 | |

## *Fractal*

**Menu:** Enable, Delete

### Fractal tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Orders*** | | | |
| Start | expression slider | 0 to 25 | |
| Levels | expression slider | 1 to 25 | |
| ***Deposition*** | | | |
| Build | expression slider | 0.0 to 3.0 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Erode | expression slider | 0.0 to 3.0 | |
| ***Slope Bias*** | | | |
| Magn | expression slider | 0 to 3 | |
| Max | expression slider | 0.0 to 3.0 | |
| Thresh | expression slider | 0 to 1.0 | |
| Mult | expression slider | 0.0 to 1.0 | |
| ***Elevation Bias*** | | | |
| Max | expression slider | 0 to 1.0 | |
| Val | expression slider | 0 to 1.0 | |
| ***Options*** | | | |
| Square | checkbox | | |
| SmoothStep | checkbox | | |

## *Erode*

**Menu:** Enable, Delete

### Erode tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Erosion*** | | | |
| Depth | expression slider | 0.0 to 2.0 | |
| Power | expression slider | 0.1 to 4.0 | |
| Start | expression slider | -2.0 to 2.0 | |
| Range | expression slider | 0.01 to 4.0 | |
| ***Drainage*** | | | |
| Start | expression slider | 0.0 to 20.0 | |
| Orders | expression slider | 1.0 to 8.0 | |
| Atten | expression slider | 0.0 to 1.0 | |
| Delf | expression slider | 1.1 to 4.0 | |
| ***Amplitude*** | | | |
| Drainage | expression slider | 0.0 to 2.0 | |
| Erosion | expression slider | 0.0 to 2.0 | |
| ***Options*** | | | |
| Rounded | checkbox | | |
| Square | checkbox | | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| SmoothStep | checkbox | | |

# Image Editor

## 1D tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***File*** | | | |
| File | list | | |
| File | list | 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 | |
| ***Options*** | | | |
| Norm | checkbox | | |
| Invert | checkbox | | |
| Show Tmps | checkbox | | |
| ***Mix/Modulate*** | | | |
| Mix/Modulate | color slider | | |
| Mod slider | slider | 0.0 to 1.0 | |
| ***Colors*** | | | |
| Colors | color slider | | |
| Enable | checkbox | | |
| ***Color Options*** | | | |
| Clamp | checkbox | | |
| Nearest | checkbox | | |
| Reflect | checkbox | | |
| Alpha | checkbox | | |
| > | button | | |
| < | button | | |

## 2D tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***File*** | | | |
| File | list | | |
| File | list | 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 | |
| File | list | 1, 2, 4, 8, 16, 32, 64, 128, | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| | | 256, 512, 1024 | |
| File | list | Tile, Decal, Sphere, Cylinder | |
| Show Tmps | checkbox | | |
| ***Options*** | | | |
| Norm | checkbox | | |
| Invert | checkbox | | |
| Gray | checkbox | | |
| Color | checkbox | | |
| Options | list | | |
| ***Function*** | | | |
| Function | expression field | | |

## Img tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***File*** | | | |
| File | list | | |
| Tile | checkbox | | |

## Map tab

Same controls as the **Img** tab.

## HMap tab

Same controls as the **Img** tab.

## Process tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***File*** | | | |
| File | list | | |
| Grayscale | checkbox | | |
| Tile | checkbox | | |
| ***Operation*** | | | |
| Operation | list | | |
| Strength | slider | 0.0 to 2.0 | |
| Radius | slider | 1 to 20 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Iterations | slider | 1 to 500 | |

# Function editors

These dialogs open from an expression field: double-click a function token, then right-click and choose **Edit**.

## *Craters*

### Craters tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Distribution*** | | | |
| Levels | list | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | |
| Density | expression slider | 0.0 to 1.0 | |
| Distribution | list | ID0, ID1, ID2, ID3, ID4, ID5, ID6, ID7, ID8, ID9 | |
| ***Max-Delta Size*** | | | |
| Max-Delta Size | slider | 0.1 to 0.9999 | |
| Max-Delta Size | list | 100, 10, 1, 0.1, 0.01, 10-3, 10-4, 10-5, 10-6 | |
| Delta size | expression slider | 0.0 to 1.0 | |
| ***Amplitude*** | | | |
| Depth | expression slider | 0.0 to 1.0 | |
| Impact | expression slider | 0 to 1.0 | |
| ***Noise*** | | | |
| Vertical | expression slider | 0 to 1.0 | |
| Radial | expression slider | 0 to 1.0 | |
| Taper | expression slider | 0 to 1.0 | |
| Bias | expression slider | 0 to 0.9 | |
| Default | button | | |

### Shape tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Geometry*** | | | |
| Rise | expression slider | 0.0 to 10.0 | |
| Drop | expression slider | 0 to 10.0 | |
| Rim | expression slider | 0.0 to 1.0 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Floor | expression slider | 0 to 1.0 | |
| Center | expression slider | 0.0 to 1.0 | |
| Offset | expression slider | -1 to 1 | |

# Fractal

## Fractal tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Orders*** | | | |
| Start | expression slider | 0 to 25 | |
| Levels | expression slider | 1 to 25 | |
| ***Deposition*** | | | |
| Build | expression slider | 0.0 to 3.0 | |
| Erode | expression slider | 0.0 to 3.0 | |
| ***Slope Bias*** | | | |
| Magn | expression slider | 0 to 3 | |
| Max | expression slider | 0.0 to 3.0 | |
| Thresh | expression slider | 0 to 1.0 | |
| Mult | expression slider | 0.0 to 1.0 | |
| ***Elevation Bias*** | | | |
| Max | expression slider | 0 to 1.0 | |
| Val | expression slider | 0 to 1.0 | |
| ***Options*** | | | |
| Square | checkbox | | |
| SmoothStep | checkbox | | |

# Noise

## Multi-Noise tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Orders*** | | | |
| Start | slider | 0 to 25 | |
| Levels | slider | 1 to 25 | |
| Atten | slider | 0 to 1 | |
| DelF | slider | 1 to 4 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Homog | slider | -1 to 1 | |
| ***Shape*** | | | |
| Sqr | checkbox | | |
| Invt | checkbox | | |
| Uns | checkbox | | |
| Abs | checkbox | | |
| Norm | checkbox | | |
| Scale | checkbox | | |
| LOD | checkbox | | |
| ***Signal*** | | | |
| Gain | slider | 0 to 1 | |
| Offset | slider | -2.0 to 2.0 | |
| ***Modulation*** | | | |
| Smooth | slider | 0 to 1 | |
| Clamp | slider | -0.5 to 2 | |

### Advanced tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Noise*** | | | |
| RSeed | list | 0, 1, 2, 3, 4, 5 | |
| RMode | list | Vertex, Pixel | |
| Type | list | Perlin, Voronoi, Simplex | |
| ***Animation*** | | | |
| On | checkbox | | |
| Animation | list | 1, 0.1, 0.01, 0.001, 1e-4, 1e-5, 1e-6, 1e-7, 1e-8 | |
| Rate | slider | 1 to 10 | |

## *Erode*

### Erode tab

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| ***Erosion*** | | | |
| Depth | expression slider | 0.0 to 2.0 | |
| Power | expression slider | 0.1 to 4.0 | |

| Control | Type | Range or choices | Notes |
|---|---|---|---|
| Start | expression slider | 0 to 1.0 | |
| Range | expression slider | 0.01 to 1.0 | |
| ***Drainage*** | | | |
| Start | expression slider | 0.0 to 20.0 | |
| Orders | expression slider | 1.0 to 8.0 | |
| Atten | expression slider | 0.0 to 1.0 | |
| Delf | expression slider | 1.1 to 4.0 | |
| ***Amplitude*** | | | |
| Drainage | expression slider | 0.0 to 2.0 | |
| Erosion | expression slider | 0.0 to 2.0 | |
| ***Options*** | | | |
| Rounded | checkbox | | |
| Square | checkbox | | |
| SmoothStep | checkbox | | |